Full Associative Cache Mapping
Sequential Sequence: [if cache block = 4] 0,1,2,3,4,5,6,7,0,1,2,3,4,5,6,7
Block Size: 4 words/block
Cache Access Time, T_c: 1ns
Memory Access Time, T_m: 100ns

Least Recently Used (LRU):

# of Accesses	Data	Block #0	Block #1	Block #2	Block #3
1	0	0	-	-	-
2	1	0	1	-	-
3	2	0	1	2	-
4	3	0	1	2	3
5	4	4	1	2	3
6	5	4	5	2	3
7	6	4	5	6	3
8	7	4	5	6	7
9	0	0	5	6	7
10	1	0	1	6	7
11	2	0	1	2	7
12	3	0	1	2	3
13	4	4	1	2	3
14	5	4	5	2	3
15	6	4	5	6	3
16	7	4	5	6	7

AMAT = hc + (1-hc)*MP, where hc = T_c * hit_rate MP = T_c + block_size * T_m + T_c AMAT = 402 ns T_{AT} = (#$_{hits}$* block_size)*T_C + (T_c+T_m*block_size+T_c*block_size) * #$_{misses}$ T_{AT} = 6480 ns	**# of hits**: 0 **# of misses**: 16 **hit rate**: 0.00 **miss rate**: 100.0 **Block Size:** 4 words/block **Cache Access Time**, T_c: 1ns **Memory Access Time**, T_m: 100ns **Total Access Time**, T_{AT}

Sequential Sequence: [if cache block = 4] 0,1,2,3,4,5,6,7,0,1,2,3,4,5,6,7
Block Size: 4 words/block
Cache Access Time, T_c: 1ns
Memory Access Time, T_m: 100ns

Most Recently Used (MRU):

# of Accesses	Data	Block #0	Block #1	Block #2	Block #3
1	0	0	-	-	-
2	1	0	1	-	-
3	2	0	1	2	-
4	3	0	1	2	3
5	4	0	1	2	4
6	5	0	1	2	5
7	6	0	1	2	6
8	7	0	1	2	7
9	0	0	1	2	7
10	1	0	1	2	7
11	2	0	1	2	7
12	3	0	1	3	7
13	4	0	1	4	7
14	5	0	1	5	7
15	6	0	1	6	7
16	7	0	1	6	7

AMAT = hc + (1-hc)*MP, where hc = T_c * hit_rate MP = T_c + block_size * T_m + T_c AMAT = 301.75 ns T_{AT} = ($\#_{hits}$* block_size)*T_C + (T_c+T_m*block_size+T_c*block_size) * $\#_{misses}$ T_{AT} = 4876 ns	**# of hits**: 4 **# of misses**: 12 **hit rate**: 4/16 **miss rate**: 12/16 **Block Size:** 4 words/block **Cache Access Time**, T_c: 1ns **Memory Access Time**, T_m: 100ns **Total Access Time**, T_{AT}

Mid-repeat Blocks: [if cache block = 4] 0,1,2,3,1,2,3,4,5,6,7, 0,1,2,3,1,2,3,4,5,6,7
Block Size: 4 words/block
Cache Access Time, T_c: 1ns
Memory Access Time, T_m: 100ns

Least Recently Used (LRU):

# of Accesses	Data	Block #0	Block #1	Block #2	Block #3
1	0	0	-	-	-
2	1	0	1	-	-
3	2	0	1	2	-
4	3	0	1	2	3
5	1	0	1	2	3
6	2	0	1	2	3
7	3	0	1	2	3
8	4	4	1	2	3
9	5	4	5	2	3
10	6	4	5	6	3
11	7	4	5	6	7
12	0	0	5	6	7
13	1	0	1	6	7
14	2	0	1	2	7
15	3	0	1	2	3
16	1	0	1	2	3
17	2	0	1	2	3
18	3	0	1	2	3
19	4	4	1	2	3
20	5	4	5	2	3
21	6	4	5	6	3
22	7	4	5	6	7

AMAT = hc + (1-hc)*MP, where $hc = T_c$ * hit_rate $MP = T_c$ + block_size * $T_m + T_c$ AMAT = 292.64 ns $T_{AT} = (\#_{hits}$* block_size)*T_C + (T_c+T_m*block_size+T_c*block_size) * $\#_{misses}$ T_{AT} = 6508 ns	**# of hits**: 6 **# of misses**: 16 **hit rate**: 6/22 **miss rate**: 16/22 **Block Size:** 4 words/block **Cache Access Time**, T_c: 1ns **Memory Access Time**, T_m: 100ns **Total Access Time**, T_{AT}

Mid-repeat Blocks: [if cache block = 4] 0,1,2,3,1,2,3,4,5,6,7, 0,1,2,3,1,2,3,4,5,6,7
Block Size: 4 words/block
Cache Access Time, T_c: 1ns
Memory Access Time, T_m: 100ns

Most Recently Used (MRU):

# of Accesses	Data	Block #0	Block #1	Block #2	Block #3
1	0	0	-	-	-
2	1	0	1	-	-
3	2	0	1	2	-
4	3	0	1	2	3
5	1	0	1	2	3
6	2	0	1	2	3
7	3	0	1	2	3
8	4	0	1	2	4
9	5	0	1	2	5
10	6	0	1	2	6
11	7	0	1	2	7
12	0	0	1	2	7
13	1	0	1	2	7
14	2	0	1	2	7
15	3	0	1	3	7
16	1	0	1	3	7
17	2	0	2	3	7
18	3	0	1	3	7
19	4	0	1	4	7
20	5	0	1	5	7
21	6	0	1	6	7
22	7	0	1	6	7

AMAT = hc + (1-hc)*MP, where

$hc = T_c * hit\_rate$

$MP = T_c + block\_size * T_m + T_c$

AMAT = 237.95 ns

$T_{AT} = (\#_{hits} * block\_size)*T_C + (T_c+T_m*block\_size+T_c*block\_size) * \#_{misses}$

T_{AT} = 5301 ns

# of hits: 9

# of misses: 13

hit rate: 9/22

miss rate: 13/22

Block Size: 4 words/block

Cache Access Time, T_c: 1ns

Memory Access Time, T_m: 100ns

Total Access Time, T_{AT}

Random sequence: 64 main memory blocks.
Block Size: 4 words/block
Cache Access Time, T_c: 1ns
Memory Access Time, T_m: 100ns

<u>Analysis Summary</u>:

Due to the low chance of acquiring a hit in this Random Sequence test case, AMAT and T_{AT} will be analyzed as if no hits were acquired.

AMAT = hc + (1-hc)*MP, where hc = T_c * hit_rate MP = T_c + block_size * T_m + T_c AMAT = 402 ns $T_{AT} = (\#_{hits} * \text{block\_size}) * T_C + (T_c + T_m * \text{block\_size} + T_c * \text{block\_size}) * \#_{misses}$ T_{AT} = 25920 ns	**# of hits**: 0 **# of misses**: 64 **hit rate**: 0.00 **miss rate**: 100.0 **Block Size:** 4 words/block **Cache Access Time**, T_c: 1ns **Memory Access Time**, T_m: 100ns **Total Access Time**, T_{AT}

Included is a trace-log from a single run of our random memory test case.

```
MISS: Word 215 -> Loaded into Cache Block 0
MISS: Word 527 -> Loaded into Cache Block 1
MISS: Word 307 -> Loaded into Cache Block 2
MISS: Word 862 -> Loaded into Cache Block 3
MISS: Word 238 -> Word address: 238 replaced word address: 215 at Block 0
MISS: Word 235 -> Word address: 235 replaced word address: 527 at Block 1
MISS: Word 950 -> Word address: 950 replaced word address: 307 at Block 2
MISS: Word 404 -> Word address: 404 replaced word address: 862 at Block 3
MISS: Word 56 -> Word address: 56 replaced word address: 238 at Block 0
MISS: Word 429 -> Word address: 429 replaced word address: 235 at Block 1
MISS: Word 522 -> Word address: 522 replaced word address: 950 at Block 2
MISS: Word 517 -> Word address: 517 replaced word address: 404 at Block 3
MISS: Word 833 -> Word address: 833 replaced word address: 56 at Block 0
MISS: Word 740 -> Word address: 740 replaced word address: 429 at Block 1
MISS: Word 933 -> Word address: 933 replaced word address: 522 at Block 2
MISS: Word 959 -> Word address: 959 replaced word address: 517 at Block 3
MISS: Word 2 -> Word address: 2 replaced word address: 833 at Block 0
MISS: Word 849 -> Word address: 849 replaced word address: 740 at Block 1
MISS: Word 90 -> Word address: 90 replaced word address: 933 at Block 2
MISS: Word 588 -> Word address: 588 replaced word address: 959 at Block 3
MISS: Word 3 -> Word address: 3 replaced word address: 2 at Block 0
MISS: Word 316 -> Word address: 316 replaced word address: 849 at Block 1
```

```
MISS: Word 739 -> Word address: 739 replaced word address: 90 at Block 2
MISS: Word 629 -> Word address: 629 replaced word address: 588 at Block 3
MISS: Word 26 -> Word address: 26 replaced word address: 3 at Block 0
MISS: Word 157 -> Word address: 157 replaced word address: 316 at Block 1
MISS: Word 603 -> Word address: 603 replaced word address: 739 at Block 2
MISS: Word 698 -> Word address: 698 replaced word address: 629 at Block 3
MISS: Word 815 -> Word address: 815 replaced word address: 26 at Block 0
MISS: Word 487 -> Word address: 487 replaced word address: 157 at Block 1
MISS: Word 512 -> Word address: 512 replaced word address: 603 at Block 2
MISS: Word 394 -> Word address: 394 replaced word address: 698 at Block 3
MISS: Word 312 -> Word address: 312 replaced word address: 815 at Block 0
MISS: Word 421 -> Word address: 421 replaced word address: 487 at Block 1
MISS: Word 678 -> Word address: 678 replaced word address: 512 at Block 2
MISS: Word 929 -> Word address: 929 replaced word address: 394 at Block 3
MISS: Word 801 -> Word address: 801 replaced word address: 312 at Block 0
MISS: Word 621 -> Word address: 621 replaced word address: 421 at Block 1
MISS: Word 95 -> Word address: 95 replaced word address: 678 at Block 2
MISS: Word 686 -> Word address: 686 replaced word address: 929 at Block 3
MISS: Word 678 -> Word address: 678 replaced word address: 801 at Block 0
MISS: Word 427 -> Word address: 427 replaced word address: 621 at Block 1
MISS: Word 915 -> Word address: 915 replaced word address: 95 at Block 2
MISS: Word 1013 -> Word address: 1013 replaced word address: 686 at Block 3
MISS: Word 100 -> Word address: 100 replaced word address: 678 at Block 0
MISS: Word 496 -> Word address: 496 replaced word address: 427 at Block 1
MISS: Word 310 -> Word address: 310 replaced word address: 915 at Block 2
MISS: Word 697 -> Word address: 697 replaced word address: 1013 at Block 3
MISS: Word 627 -> Word address: 627 replaced word address: 100 at Block 0
MISS: Word 476 -> Word address: 476 replaced word address: 496 at Block 1
MISS: Word 809 -> Word address: 809 replaced word address: 310 at Block 2
MISS: Word 93 -> Word address: 93 replaced word address: 697 at Block 3
MISS: Word 832 -> Word address: 832 replaced word address: 627 at Block 0
MISS: Word 133 -> Word address: 133 replaced word address: 476 at Block 1
MISS: Word 945 -> Word address: 945 replaced word address: 809 at Block 2
MISS: Word 535 -> Word address: 535 replaced word address: 93 at Block 3
MISS: Word 621 -> Word address: 621 replaced word address: 832 at Block 0
MISS: Word 158 -> Word address: 158 replaced word address: 133 at Block 1
MISS: Word 505 -> Word address: 505 replaced word address: 945 at Block 2
MISS: Word 69 -> Word address: 69 replaced word address: 535 at Block 3
MISS: Word 104 -> Word address: 104 replaced word address: 621 at Block 0
MISS: Word 864 -> Word address: 864 replaced word address: 158 at Block 1
MISS: Word 399 -> Word address: 399 replaced word address: 505 at Block 2
MISS: Word 767 -> Word address: 767 replaced word address: 69 at Block 3
```